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FFIEC

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Systems

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HANDBOOK

Table of Contents

Introduction	1
Interbank Payment and Messaging Systems	2
Fedwire and Clearing House Interbank Payments System (CHIPS)	2
Fedwire Funds Service	3
CHIPS	4
Other Clearinghouse, Settlement, and Messaging Systems	5
National Settlement Service (NSS)	5
Society for Worldwide Interbank Financial Telecommunication (SWIFT)	6
Telex-based Messaging Systems	7
Continuous Linked Settlement (CLS) Bank	7
Securities Settlement Systems	8
U.S. Government Securities	8
Fixed Income Clearing Corporation (FICC)	9
Fedwire Securities Service	10
Corporate and Municipal Securities	11
National Securities Clearing Corporation (NSCC)	11
Depository Trust Company (DTC)	12
Intrabank Payment and Messaging Systems	12
Internally Developed and Off-The-Shelf Funds Transfer Systems	12
Payment Messaging Systems	13
In-house Terminals	14
Non-automated Payment Order Origination	14
Funds Transfer Operations (Wire Room)	15
Computer and Network Operations Supporting Funds Transfer	16
Wholesale Payment Systems Risk Management	17
Payments System Risk (PSR) Policy	17
Reputation Risk	18

Strategic Risk	18
Credit Risk	18
Customer Daylight Overdrafts	18
Settlement Risk	19
Liquidity Risk	21
Legal (Compliance) Risk	21
Operational (Transaction) Risk	24
Internal and Operational Controls	24
Audit	26
Information Security	26
Business Continuity Planning (BCP)	27
Vendor and Third-Party Management	27
Appendix A: Examination Procedures	A-1
Appendix B: Glossary	B-1
Appendix C: Laws, Regulations and Guidance	C-1
Appendix D: Legal Framework for Interbank Payment Systems	D-1
Appendix E: Federal Reserve Board Payment System Risk Policy: Daylight Overdrafts	E-1
Appendix F: Payment System Resiliency	F-1

Introduction

This Wholesale Payment Systems Booklet (Booklet) is one of several that comprise the Federal Financial Institutions Examination Council (FFIEC) Information Technology (IT) Examination Handbook (IT Handbook). It provides guidance to examiners and financial institution management regarding the risks and risk-management practices when originating and transmitting large-value payments. It replaces and rescinds Chapter 18 of the 1996 Information Systems Examination Handbook.

This booklet is organized into the following four sections describing the various aspects of wholesale payment systems, followed by examination procedures, a glossary, a discussion of the legal framework for interbank payment systems, a discussion of the Federal Reserve Board's Payments System Risk (PSR) Policy, and a discussion of the "Interagency Paper on Sound Practices to Strengthen the Resiliency of the U.S. Financial System." Management action summaries are included in each of the four sections providing a snapshot of the risks and risk management practices described in the text.

- **Interbank Payment and Messaging Systems** - The first section includes a discussion of interbank funds transfer and messaging systems, and interbank clearing and settlement systems. This section concludes with a discussion of Internet-based wholesale payment systems.
- **Intrabank Payment and Messaging Systems** - The second section provides an overview of the funds transfer and messaging systems employed by typical financial institutions.
- **Securities Settlement Systems** - The third section provides an overview of the major securities markets and securities settlement systems. Securities settlement generates a large volume of wholesale payments typically processed by financial institutions. The information presented in this section describes these settlement processes. While securities settlement generally involves both the delivery of a security and the corresponding payment, this section focuses on the wholesale payments generated by these transactions, and does not cover the complete range of activities associated with securities processing, including the transfer and delivery of securities, custodial arrangements, and related trading activities.^[1]
- **Wholesale Payment Systems Risk Management** - The fourth section describes the risks associated with wholesale payment systems, using various risk categories, including reputation, strategic, credit, liquidity, legal/compliance, and operational/transaction risk. This section also describes the risk management measures management should establish to mitigate the risks described.

The booklet references a number of other IT Handbook booklets, including the "Information Security Booklet," "Business Continuity Planning Booklet," "Outsourcing Technology Services Booklet," and the "FedLine Booklet."^[2] In addition to describing the information technology risks and controls, the booklet also describes certain credit and liquidity risks that may be present when conducting wholesale payment services. A full review of a financial institution's wholesale payment system environment may require the

use of examiners with experience in credit, liquidity, and compliance issues and additional agency-specific examination procedures.

Examiners should use the examination procedures (detailed in Appendix A) for reviewing risks in wholesale payment systems. These procedures address services and products of varied complexity, and examiners should adjust the procedures for the scope of the examination and the size and risk profile of the institution. Examiners may use the procedures independently or in combination with procedures from other IT Handbook booklets and agency specific handbooks and guidance.

For financial institutions whose primary method for originating, transmitting, and receiving large-value payment orders is the use of the FedLine Funds Transfer (FT) application, examiners should use the "FedLine Booklet" to assess the wholesale payment systems and funds transfer environment. In addition, examiners should also review the "Legal Framework" and "PSR" discussions in Appendices D and E, respectively, if an expanded evaluation of the funds transfer operation is necessary.

Interbank Payment and Messaging Systems

Fedwire and Clearing House Interbank Payments System (CHIPS)

In the United States, payment and securities settlement systems consist of numerous financial intermediaries, financial services firms, and non-bank businesses that create, distribute, and process large-value payments. The bulk of the dollar value of these payments are processed electronically and are generally used to purchase, sell, or finance securities transactions; disburse or repay loans; settle real estate transactions; and make large-value, time-critical payments, such as payments for the settlement of interbank purchases and sales of federal funds, settlement of foreign exchange transactions, or other financial market transactions.

There are two primary networks for interbank, or large-value, domestic, funds transfer payment orders. The first, Fedwire® Funds Service, is operated by the Federal Reserve Banks, and is an important participant in providing interbank payment services as well as safekeeping and transfer services for U.S. government and agency securities, and mortgage-backed securities. ^[3] Funds Service and the Federal Reserve's National Settlement Service (NSS) are critical components used in other payment systems' settlement processes. The Clearing House Interbank Payments Company L.L.C. (CHIP Co.) operates the second, the Clearing House Interbank Payments System (CHIPS). ^[4]

Processing large-value funds transfers involves two key elements: clearing and settlement. Clearing is the transfer and confirmation of information between the payer (sending financial institution) and payee (receiving financial institution). Settlement is the actual transfer of funds between the payer's financial institution and the payee's financial institution. Settlement discharges the obligation of the payer financial institution to the payee financial institution with respect to the payment order. Final settlement is irrevocable and unconditional. The finality of the payment is determined by that system's rules and applicable law.

In general, payment messages may be credit transfers or debit transfers. Most large-value funds transfer systems are credit transfer systems in which both payment

messages and funds move from the payer financial institution to the payee financial institution. An institution initiates a funds transfer by transmitting a payment order (a message that requests the transfer of funds to the payee). Payment order processing follows the predefined rules and operating procedures of the large-value payment system used. Typically, large-value payment system operating procedures include identification, reconciliation, and confirmation procedures necessary to process the payment orders. In some systems, financial institutions may contract with one or more third parties to help perform clearing and settlement activities on behalf of the institution.

The legal framework governing payment activity and the regulatory structure for financial institutions that provide payment services is complex. ^[5] There are rules for large-value payments that are distinct from retail payments. Large-value funds transfer systems differ from retail electronic funds transfer (EFT) systems, which generally handle a large volume of low value payments including automated clearinghouse (ACH) and debit and credit card transactions at the point of sale.

Fedwire Funds Service

Fedwire Funds Service is a real-time gross settlement system (RTGS) enabling participants to transmit and receive payment orders between each other and on behalf of their customers. Real-time gross settlement means that the clearing and settlement of each transaction occurs continuously during the processing day. Payment to the receiving participant (payee) over Fedwire Funds Service is final and irrevocable when the Federal Reserve Bank either credits the amount of the payment order to the receiving participant's Federal Reserve Bank reserve account or sends notice to the receiving participant, whichever is earlier.

Fedwire Funds Service participants must maintain an account with a Federal Reserve Bank. Because of this requirement, non-financial organizations are not permitted direct access to Fedwire Funds Service, although these entities may use these services indirectly as customers of deposit-taking financial institutions. Subject to the Federal Reserve Bank's and the Federal Reserve Board's risk reduction policies, where applicable, entities authorized by law, regulation, policy, or agreement to be participants include depository institutions, agencies and branches of foreign banks, member banks of the Federal Reserve System, the Treasury and any entity specifically authorized by federal statute to use the Reserve Banks as fiscal agents or depositories, entities designated by the Secretary of the Treasury, foreign central banks, foreign monetary authorities, foreign governments, and certain international organizations; and any other entities authorized by a Federal Reserve Bank to use Fedwire Funds or Security Services. See Appendix E for a discussion of the Federal Reserve Board's PSR Policy. Certain payment and securities settlement systems, such as CHIPS and CLS, also rely upon Fedwire Funds Service to allow participants or their correspondents to provide necessary funding. CHIPS and CLS also use Fedwire Funds Service for settlement services by establishing zero-balance settlement accounts to settle clearinghouse participant obligations. The Federal Reserve Bank requires participants in a net debit position to make a Fedwire Funds Service funds transfer payment to the settlement

account. It then pays participants in a net credit position by means of a Fedwire Funds Service funds transfer from that settlement account. Once the Federal Reserve Bank processes these funds transfers, they are final and irrevocable.

Financial institutions sending a Fedwire Funds Service payment order irrevocably authorize their Federal Reserve Bank to debit (charge) their Federal Reserve account for the transfer amount and to give credit in the same amount to the payee. Only the originating financial institution can have funds removed from its Federal Reserve account using the Fedwire Funds Service. Depository institutions that maintain a reserve or clearing account with a Federal Reserve Bank may use Fedwire Funds Service to send payments to, or receive payments from, other account holders directly. Once the Federal Reserve Bank credits the receiving institution's account, it will not reverse the transaction at the request of the originating institution.

Financial institutions may access the Fedwire Funds Service via high-speed direct computer interface (CI), FedLine, or with off-line telephone connectivity with a Federal Reserve Bank. Financial institutions may also access certain Fedwire Funds Service inquiry information via FedLine for the Web. By year-end 2004, customers should be able to initiate Fedwire Funds Service and Fedwire Securities Service transactions through the web via FedLine Advantage. On-line participants, using either a CI or FedLine PC connection to Fedwire Funds Service, require no manual processing by the Federal Reserve Banks. Off-line participants provide funds transfer instructions to one of two Federal Reserve Bank customer support sites by telephone, and after authenticating the participant, the Federal Reserve Bank enters the transfer instruction into the Fedwire Funds Service system for execution. The manual processing required for off-line requests makes them more costly and suitable only for institutions processing a small number of funds transfer payment orders.

The Federal Reserve Bank's FedLine for the Web currently offers access to low-risk Federal Reserve Bank financial services. FedLine Advantage, which should begin a graduated rollout by year-end 2004, will allow depository institutions access to additional Federal Reserve financial services, including the Fedwire Funds Services and the Fedwire Securities Service, via a secure Internet Protocol (IP) gateway to Federal Reserve Bank financial services. Residing on a secure Web server, FedLine Advantage will be accessible to customer financial institutions with authenticated credentials using digital certificates.

CHIPS

CHIPS is a privately operated, real-time, multilateral, payments system typically used for large dollar payments. CHIPS is owned by financial institutions, and any banking organization with a regulated U.S. presence may become an owner and participate in the network. The payments transferred over CHIPS are often related to international interbank transactions, including the dollar payments resulting from foreign currency transactions (such as spot and currency swap contracts) and Euro placements and returns. Payment orders are also sent over CHIPS for the purpose of adjusting correspondent balances and making payments associated with commercial transactions, bank loans, and securities transactions.

Since January 2001, CHIPS has been a real-time final settlement system that continuously matches, nets and settles payment orders. This system provides real-time finality for all payment orders released by CHIPS from the CHIPS queue. To achieve real-time finality, payment orders are settled on the books of CHIPS against participants'

positive positions, simultaneously offset by incoming payment orders, or both. This process is dependant on up to two rounds of required prefunding.

To facilitate this prefunding, CHIP Co. members jointly maintain a pre-funded balance account (CHIPS account) on the books of the Federal Reserve Bank of New York. Under the real-time finality arrangement, each CHIPS participant has a pre-established opening position (or initial prefunding) requirement, which, once funded via a Fedwire Funds Service funds transfer to the CHIPS account, is used to settle payment orders throughout the day.^[6] A participant cannot send or receive CHIPS payment orders until it transfers its opening position requirement to the CHIPS account. Opening position requirements can be transferred into the CHIPS account any time after the opening of CHIPS and Fedwire Funds Service at 9:00 p.m. Eastern Time. However, all participants must transfer their requirement no later than 9:00 a.m. Eastern Time.

During the operating day, participants submit payment orders to a centralized queue maintained by CHIPS. Participants may remove payment orders from the queue at any time prior to the daily cutoff time for the system (5:00 p.m. Eastern Time). When an opportunity for settlement involving one, two or more payment orders is found, the system releases the relevant payment orders from the central queue and simultaneously marks the CHIPS records to reflect the associated debits and credits to the relevant participant's positions. Debits and credits to the current position are reflected only in CHIPS records and are not recorded on the books of the Federal Reserve Bank of New York. Under New York law and CHIPS Rules, payments orders are finally settled at the time of release from the central CHIPS queue.

This process, however, typically will be unable to settle all queued messages. Soon after 5:00 p.m. Eastern Time, CHIPS tallies any unreleased payment orders remaining in the queue on a multilateral net basis. The resulting net position for each participant is provisionally combined with that participant's current position (which is always zero or positive) to calculate the participant's final net position; if that position is negative, it is the participant's "final position requirement."

Each participant with a final position requirement must transfer, via Fedwire Funds Service, this second round of prefunding to the CHIPS account. These requirements, when delivered, are credited to participants' balances. Once all of the Fedwire Funds Service funds transfers have been received, CHIPS is able to release and settle all remaining payment orders. After completion of this process, CHIPS transfers to those participants who have any balances remaining the full amount of those positions, reducing the amount of funds in the CHIPS account to zero by the end of the day. In the event that less than all final position requirements are received, CHIPS settles as many payments as possible, subject to the positive balance requirement, and deletes any remaining messages from the queue. Participants with deleted messages are informed of which messages were not settled, and may choose, but are in no way required, to settle such messages over Fedwire Funds Service.

Other Clearinghouse, Settlement, and Messaging Systems

National Settlement Service (NSS)

NSS is a multilateral settlement service owned and operated by the Federal Reserve Banks. It allows participants in private clearing arrangements to settle their net

obligations with same-day finality using participant's reserve or clearing account balances maintained at the Federal Reserve Banks. NSS participants include local check clearinghouse associations, automated clearinghouse (ACH) networks, credit card processors, and automated teller machine (ATM) networks.

To use NSS, a settlement agent transmits a settlement file electronically to a Federal Reserve Bank. The file contains a listing of the participants, the settlers (either the participant itself or the participant's correspondent), and the dollar amount of the debit or credit to be posted to the settler's account. If validity checks are satisfied, the Federal Reserve Bank accepts the file for processing and sends an acknowledgment to the agent. The Federal Reserve Bank accepts NSS files for processing and settlement between 8:30 a.m. and 5:30 p.m. Eastern Time. If an institution submits files earlier than 8:30 a.m. Eastern Time, they enter a queue for processing beginning at 8:30 a.m. Eastern Time. The NSS checks each debit balance on the settlement file against the account balance and intraday credit available to the settlers. The system may reject debit balances if a settler does not have a sufficient balance or sufficient intraday credit to cover the debit. Once it has posted all debit entries on the settlement file, NSS posts the credit balances. All transactions are final and irrevocable once the NSS posts them. The settlement for a file is complete when all credits have been posted. The NSS then sends an acknowledgment message to the settlement agent.

NSS offers payment finality similar to that of the Fedwire Funds Service and provides an automated mechanism for submitting settlement files to the Federal Reserve Banks. NSS improves operational efficiency and reduces settlement risk to participants by providing settlement finality on settlement day. It also enables the Federal Reserve Banks to manage and limit risk by incorporating risk controls similar to those used in the Fedwire Funds Service. NSS can settle transactions across Federal Reserve Districts or within a single Federal Reserve District.

Society for Worldwide Interbank Financial Telecommunication (SWIFT)

International funds transfer operates differently from domestic large-value funds transfer. While the SWIFT operates as a messaging system, transmitting instructions to move funds, the domestic systems discussed above accomplish the actual funds movement.

SWIFT is an industry-owned cooperative, which supplies secure, standardized messaging services and interface software to 7,500 financial institutions in 199 countries. The SWIFT community includes a variety of financial services firms, including banks, broker/dealers, and investment managers, as well as their market infrastructures in payments, securities, treasury, and trade. The SWIFT core application is the FIN (Financial Transaction) application, which provides a store and forward financial messaging service accessed over X.25 network connections.^[7] The system now uses bilateral key security and provides message validation to ensure messages are formatted according to SWIFT standards.

SWIFT messages may be used to transmit payment instructions for both domestic and international financial transactions. Financial institutions generating a high volume of funds transfer activity typically establish direct SWIFT connectivity with their internal, intrabank funds transfer system. The SWIFT network controls the integrity of the messages once properly entered at the point of origination. Thus there is no requirement for the receiver to re-verify payment orders. As a result, payment instructions pass through the system without human intervention unless programmed conditions (e.g., overdraft limit excesses) or error messages occur.

In 2002, SWIFT began migrating its core FIN application from an X.25 network to SWIFTNet, an IP-based network using the Secure Internet Protocol Network (SIPN). The introduction of IP-based technologies, that uses both bilateral keys and public keys, will allow SWIFT to expand its services. Payment-related business solutions currently being developed include cash reporting, bulk payments processing, and securities reporting. To promote the use of SWIFTNet and XML (Extensible Markup Language) based cash reporting tools among major cash clearing institutions and their correspondents, a working group has been set up to design the industry solution, including query/response standards and a rule book. In bulk payments, SWIFT developed a new XML-based message standard that was introduced in 2002.

Telex-based Messaging Systems

Several private telecommunications companies offer worldwide or interconnected services that provide a printed record of each transmitted message. Financial institutions that do not have access to SWIFT use Telex-based proprietary systems. Financial institutions access Telex-based systems via a dial-up or a dedicated line with dedicated printers. Some systems are computer monitored 24 hours a day, seven days a week and are fully redundant with automatic switch over and recovery capability.

Telex systems do not include built-in security features. Financial institutions that use Telex must exchange security codes. Typically, senders number messages sent to another institution to provide better audit controls. The sending institution is responsible for incorporating a test key in all instructions to a receiver to execute a payment order. The receiver is responsible for safekeeping the unique test code keys of each sender and decoding each test message. This function should be clearly separated from the Telex operating area and funds payment function. Presently, only a few institutions employ fully automated interface of Telex with their funds payment systems.

Continuous Linked Settlement (CLS) Bank

CLS Bank is a private sector special purpose bank that uses the CLS system to settle payments associated with a foreign exchange (FX) transaction simultaneously.

CLS is designed to eliminate the risk occurring when each leg of certain FX transactions is settled separately, i.e., the payment is made and the corresponding payment is not received. CLS Bank settles payment instructions in the following currencies: Australian Dollar, Canadian Dollar, Danish Krone, Euro, Japanese Yen, Norwegian Krone, Singapore Dollar, Swedish Krona, Swiss Franc, GB Pound, and U.S. Dollar, and is expected to add more currencies over time.

Customers may join CLS Bank as either a Settlement Member or User Member. Settlement members have accounts with CLS Bank and must sponsor User members that do not have such accounts. Any Settlement member or User member may submit payment instructions for settlement processing, but the sponsoring Settlement member must authorize each instruction a User member submits. Non-members may settle their FX payments through private arrangements with members who will submit the settlement processing instructions to the service. Such members are responsible to CLS Bank as principals with respect to such payment instructions.